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Different bees as vectors for entomovectoring with enhanced pollination and crop protection control: current practices, use cases and critical view on transport

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Summary

Honeybees, bumblebees and other insects have been used commercially for pollination for many years, and microbial biocontrol agents have also been widely used in pest control. Pollinators and formulations of microbial pest-control agents are routinely transported internationally on a large scale. A novel approach has been developed to use bees as vectors of microbial agents by inoculating the surface of the pollinators using dispensers in modified hives. This innovation extends the market for these products and results in better yields. A successful entomovector system requires selecting the vector pollinator most appropriate for the crop and location, based on various criteria, in combination with a registered microbial agent. Currently, pollinators and microbial agents are packed separately and combined at the point of use. Local sourcing of the pollinator in the system reduces the need for long-distance shipping of these live insects and may improve efficiency due to local adaptation; however, it will delay use and benefits of the system until research at each site/country catches up with the work already conducted in a few countries. In the meantime, clear guidance for innovative systems employing live insects could support the promising increase in food production.

Keywords

Bees – Crop protection – Entomovectoring – Innovative biological systems – Microbial control – Pollination.

Introduction

Both pollinators and biocontrol agents (BCAs) are broadly commercialised internationally. In 2015, the combined global area treated with invertebrate and microbial agents amounted to more than 30 million hectares [1]. Commercial pollination by bumblebees alone was already estimated to be a € 55 million industry in 2004 with over a million colonies produced worldwide [2]. In order to further optimise pollination and application of microbial BCAs, the two international functions have been combined through entomovectoring.

The entomovectoring system is a good example of humans copying nature. It has long been established that insects such as bees are able to transfer pollen and other microparticles such as bacterial cells and fungal spores from flower to flower, as demonstrated for *Erwinia amylovora* [3]. Humans have adopted this concept and used it for commercial pollination and, more recently, for the transport of microbial BCAs. Hokkanen and Menzler-Hokkanen [4] introduced the technical term 'entomovector technology', meaning the use of vectors such as insects for transporting microbial BCAs for plant protection. The entomovectoring system has the potential to increase the market further, but it also introduces new concepts that need to be understood when applying guidance on international risk assessment and shipping. For example, during transport, the vectors are packed separately from the BCA. The shipment itself does not differ from or require extra management compared to standard pollinator transport. Misperceptions about the purpose and operation may occur during shipping, however, which is why detailed descriptions are necessary.

As early as 1992, the honeybee *Apis mellifera* was used to transfer the fungus *Gliocladium roseum*, a commercially available microbial BCA, for the protection of strawberry

plants against the notorious plant pathogen *Botrytis cinerea* [5].

The entomovectoring system makes use of a unique combination to simultaneously provide pollination and a protective service by using pollinators such as the honeybee for the transport of microbial BCAs. As a result, the grower benefits from increased yields due to an increased seed set and protection against plant pathogens. There are other benefits from using entomovectoring instead of spraying chemical control agents. Aside from avoiding chemicals that can have adverse effects on the environment and human health [6], the entomovectoring system can reduce water and electricity requirements associated with glasshouse production, resulting in a lower carbon footprint, an outcome that fits within the framework of the European Green Deal (https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en) [7].

Entomovectoring reduces costs and labour hours and decreases non-target organism exposure [8]. In terms of hive management, the greatest requirement, apart from installation, is to refill the dispenser with BCA product, which takes approximately 15 to 30 seconds per hive, either twice a week for bumblebees or every ten days for honeybees. Commercially available colonies require placement on a small stand, protection from rain (most of the hives have a paper-based coating) and access to pollen and nectar. Honeybees require year-round maintenance to be ready for application of microbial BCAs in the desired period. In addition, using pollinating insects for the delivery of the microbial BCAs ensures that the BCAs will be disseminated directly onto the target location, that is, on the flowers from the moment they open [9]. This is important as these are often the target organs for harmful plant pathogens [10, 11]. Furthermore, less microbial BCA is wasted because during spraying a significant amount of the product ends up on the ground.

The economic benefits of implementing entomovectoring with the honeybee and *Trichoderma harzianum* for the control of *Botrytis cinerea* in field strawberry plants were estimated in Colombia [12]. Pollination resulted in an extra profit of US$ 2,987/ha/year. The decrease in application of pesticides saved US$ 1,405/ha/year, and a 1.88% reduction in losses due to *Botrytis cinerea* compared to commercial treatment saved US$ 872/ha/year. Overall, the entomovectoring technology brought in utility close to US$ 3,710/ha/year, which is the sum of the three values above minus the costs for the implementation of the technology (US$ 1,554/ha/year) [12].

This review will provide a brief recap of how entomovectoring works, especially in relation to transport. Some historic cases are presented that indicate the need to use local species in entomovectoring to protect pollinator biodiversity. This article aims to share information with stakeholders on the new concept of entomovectoring and its advantages and disadvantages, together with a critical view on transport of pollinators for the purpose of entomovector systems, demonstrating the need for clarity around any related regulations on export and import.

Entomovectoring

Principles of entomovectoring

Entomovectoring involves an interplay of several parts (Figure 1). A certain pollinator, referred to as the vector, must deliver a microbial BCA product to the crop under protection in a way that is safe for humans, the vector, and the environment. To ensure the vector contains a sufficient number of colony-forming units of the microbial BCA, it must be loaded using a dispenser. For this purpose, the vector walks through a device filled with the BCA formulation. The dispenser must provide a way to load the vector safely and sufficiently without significantly altering the physiological behaviour of the vector or compromising its health and safety [7]. To ensure that the BCA has no adverse effects on humans, vectors, or the environment, risk assessments covering safety must be completed and product registration must be in place [14].

The vector

Proper application of the microbial BCA to the flowers contributes significantly to the successful operation of the entomovectoring system [11]. Matching the appropriate vector with the plant under protection is thus highly important. The choice for the most suitable vector is dependent on several factors, including the crop species, the

Figure 1

Schematic overview of the factors important for successful entomovectoring and their interplay [11, 13]

crop-visitation rate of the vector, the flying conditions for the vector and the vector's intrinsic capacity to disseminate microbial BCAs [11].

The number of insect species that have been used in the entomovectoring system is limited. The bottleneck is in linking the commercial availability of the pollinator with the blooming period of the plant under protection as the pollinator life cycle is not always related to its blooming period [11]. This is also why honeybees and bumblebees, both commercially available throughout the year, are very much in demand for use in the entomovectoring system and have been the vector in many scientific reports investigating entomovectoring [13]. For instance, *Apis* and *Bombus* workers differ in pollen deposition and removal because of different foraging behaviour. Examples of crop–pollinator links were found for numerous species of the Asteraceae family that proved to be particularly attractive to honeybees and wild bees [15]. Members of the Brassicaceae family were mostly disliked by solitary bees and bumblebees but were visited frequently by honeybees. Bumblebees were notably present on plants of the Leguminosae family [15]. Apart from honeybees and bumblebees, solitary bees such as *Megachile rotundata, Osmia bicornis, Osmia cornuta* and *Osmia lignaria* have been used as vectors as well [16]. Plans are being made for the use of stingless bees (*Meliponini*) for entomovectoring in Latin America. The stingless bee species *Nannotrigona perilampoides* has pollinated tomatoes successfully [17]. However, more research and new multiplication methods are needed for many stingless bee species before they can be used commercially [2].

Two main pollinators in entomovectoring: honeybees and bumblebees

Commercially, honeybees (*Apis mellifera*) have been used in North America, Europe, South America and Asia [13, 18, 19]. Bumblebees have been successfully deployed in North America (*Bombus impatiens*), Europe (*Bombus terrestris*) and the Republic of Korea (*Bombus terrestris*) [9, 13, 18]. The conditions under which the vectors must fly are decisive in determining which vector will generate the best result for pollination. This also applies to the choice between bumblebees and honeybees. For example, it is known that honeybee workers do not fly, or fly poorly, under rainy conditions and that rain affects their ability to disseminate BCAs [11, 20]. Bumblebees, which are described as bad-weather foragers [20], perform better under rainy conditions. In addition, honeybees have been described as less suitable for entomovectoring in greenhouse crops [21]. However, bumblebees can better tolerate higher temperature extremes and are also more resistant to temperature fluctuations [20]. This is why bumblebees are the most used pollinators inside greenhouses [22]. Nevertheless, bumblebees are also suitable for entomovectoring in open field conditions [11].

Another important advantage that bumblebees have over the honeybee is the fact that they perform buzz pollination [20, 21]. This is a highly efficient pollination technique in which the flower stamens are shaken intensively by the vibrating wings of the insect, resulting in the formation of a cloud of pollen grains [23]. Using this technique, bumblebees can pollinate flowers more efficiently than honeybees [22].

Case studies with the honeybee, *Apis mellifera*

The first study on entomovectoring was conducted by Peng *et al.* [5] for the protection of strawberry plants against *Botrytis cinerea* with *Apis mellifera* and the microbial BCA *Gliocladium roseum.* This system was able to suppress *Botrytis cinerea* on the flowers in the greenhouse and in the open field, except when the weather conditions were bad due to the resulting reduced foraging activity.

Honeybees and bumblebees were used for entomovectoring the fungus *Trichoderma harzianum* on strawberry plants in open fields for the control of *Botrytis cinerea* [18]. This entomovectoring system resulted in better control of *Botrytis cinerea* than conventional spraying. The more targeted and continuous application of the BCA at flower opening – in comparison to spraying, which occurred before most flowers had opened – resulted in flowers being immediately protected at opening: the unprotected time is significantly less compared to spraying. Furthermore, entomovectoring resulted in on average 22% more seeds, with a 26% to 40% (by weight) higher yield as a result [18]. Entomovectoring with *Trichoderma harzianum* resulted in a level of control and a yield that were as good as or better than those achieved with commercial fungicides plus pollination [18].

In another study [7], experiments were carried out on open field strawberry plants in Finland. During these experiments, honeybees loaded with Prestop-Mix, with the active component *Gliocladium catenulatum,* increased the yield by 58%, due owing to enhanced pollination and avoidance of fungicides with potential negative effects on the seed set.

Case studies with the bumblebees *Bombus terrestris* and *Bombus impatiens*

Kapongo *et al.* [19] demonstrated how *Clonostachys rosea* transported by *Bombus impatiens* suppressed *Botrytis cinerea* by 57% and 46% in the flowers and leaves of the tomato plant, respectively, and by 58.9% and 46.8%, respectively, in the flowers and leaves of the sweet pepper. Mommaerts *et al.* [24] investigated entomovectoring with Prestop-Mix and the vector *Bombus terrestris* to combat *Botrytis cinerea* on the strawberry plant under greenhouse conditions. They observed that the use of this BCA and vector could provide a significantly higher yield (2 to 2.5 times higher) compared to the control group without BCAs, even under conditions that were considered ideal for the fungus.

Several companies have developed business models based on entomovectoring technology. Examples are Biobest (Belgium), Bee Vectoring Technology (BVT, Canada) and Assatek (Finland). An example of a commercially available product is the Flying Doctors hive (Figure 2). This is a patented nest sold by Biobest and used especially for vectoring with *Bombus terrestris.* The dispenser is based on the Mommaerts dispenser [25].

Other potential pollinators in entomovectoring: solitary bees *Osmia* species

Substantial research has examined the pollination potential of native solitary bees [16]. Wild bees can maintain significant populations, providing ecosystems with constant background pollination. Thus far, several manageable native solitary bees have been identified. *Peponapis pruinose*, for example, pollinates *Cucurbita pepo* very efficiently. Sampson and Cane [26] successfully bred a new adaptable pollinator, *Osmia ribifloris,* which was used in captivity to pollinate rabbiteye blueberry. The authors found that this insect pollinated the blueberries as efficiently as the honeybee.

Currently, few solitary bee species are commercially available. This is due to the limited possibility to develop mass-reared populations of pollinators. Difficulties include the number of reproductive cycles, the duration of the diapause and the adaptability to artificial nesting sites [16]. In North America, *Megachile rotundata, Osmia lignaria* and *Nomia melanderi* are available on a larger scale. *Osmia cornifrons* has been used for over 70 years in Japan for apple pollination. In Europe, *Osmia rufa* and *Osmia cornuta* are bred for orchard pollination [16]. The genus *Osmia* consists of 300 species, most of whose life cycles are related to the blooming period of the Rosaceae family. Members of the *Osmia* genus strongly prefer Rosaceae pollen over other flowers [16]. *Osmia* species have been used on several occasions for crop pollination. *Osmia cornuta* has been found to pollinate apple flowers more efficiently than *Apis mellifera* [27]. Solitary bees can move more easily from tree to tree compared to honeybees, which makes them more suitable for pollination in orchards [16]. Furthermore, solitary bees are better at maintaining a constant population level in orchards compared to honeybees and bumblebees, which makes them reliable pollinators [28].

Case studies with solitary bees

Maccagnani *et al.* [29] studied the efficiency of two pollinators, *Apis mellifera* and *Osmia cornuta*, as vectors for the BCA *Bacillus subtilis* BD 170 on apple cultivar Golden Delicious for control of *Erwinia amylovora,* the causal agent of fire blight, under net-screened tunnels. The authors studied secondary colonisation through transport from one flower to another. They found that, in 50% of cases, the BCA carry-over for *Osmia cornuta* was 10 to 100 times higher than that of *Apis mellifera*. The solitary bee *Osmia cornuta* achieved higher percentages of inoculated flowers in addition to a higher proportion of flowers with a large number of BCA cells. These results are in line with the higher pollinating potential of solitary bees on crops of the Rosaceae family [29].

Maccagnani *et al.* [30] investigated the suitability of *Osmia cornuta* and *Apis mellifera* to vector *Bacillus subtilis* BS-F4 for

Figure 2

(a) Flying Doctors hive from Biobest (Westerlo, Belgium) (https://www.biobestgroup.com/nl/biobest/producten/hommelbestuiving-4456/)
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(b) Loading tray for Flying Doctors hive filled with biocontrol agent formulation
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(c) Bumblebee hive for enhanced pollination in strawberry plants under protection
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the control of fire blight on pear cultivar Abbé Fétel. Overall, flowers visited by *Osmia cornuta* showed a greater number of BCA cells than flowers visited by *Apis mellifera*.

Transport of vectors: impact of international import and export

Current status

As important pollinators, bumblebees (*Bombus* spp.) have been bred and used for pollination since the late 1980s. In 1987, bumblebees were first used in Belgium for the commercial pollination of tomato plants [2]. In 2006, Velthuis and van Doorn [2] indicated commercial pollination by bumblebees as a € 55 million industry, with over a million colonies produced worldwide in 2004. Companies such as Biobest produce bumblebees for several areas of the world (Table I).

Potential risks in importing exotic vectors

The increasing commercialisation and the associated anthropogenic movement of commercial pollinators in and outside their natural ranges can have an impact on native bee species [31]. Moreover, commercial agricultural populations can impact local bee fauna through several mechanisms [32]. For example, they are said to spread harmful pathogens to native wild bee populations. In this sense, commercial populations act as pathogen reservoirs that can contaminate wild populations. Several commercial populations of bees have been reported to contain elevated parasite loads [31]. Furthermore, in many cases the foraging activity of invasive bumblebee species overlaps with that of native bumblebee species. As such, the invasive species can replace the native species through competitive exclusion [33]. Competition for both floral resources and nesting sites can have negative effects on the local bee fauna [32]. Finally, interspecific mating between invasive and native bumblebee species can have a detrimental impact on the native bee species [32].

Table I

Bumblebee species produced by Biobest for several locations (from https://www.biobestgroup.com/en/biobest/pollination/things-to-know-about-bumblebees-7052/species-6674/)

Species	Production	Availability
Bombus terrestris	Belgium, Turkey, China	Europe, Africa, Asia, Chile
Bombus terrestris audax	Belgium	United Kingdom
Bombus canariensis	Belgium	Canary Islands
Bombus ignitus	Belgium	Japan
Bombus impatiens	Canada, Mexico	North, Central and South America
Bombus atratus	Argentina	South America
Bombus huntii	Canada	Western Canada

The global trade of bees facilitates the introduction of potentially invasive species into new environments (see also Goka [34], this issue). Regulations are often still too regional to help in solving this global problem. For example, Norway has banned the introduction of commercial bumblebees and uses only species that are local or grown in the country to protect its native species. Estonia, on the other hand, has no national risk assessment concerning pollinator introduction. The reasoning is that since the use of pollinator services or entomovectoring is not common in Estonia, there has not been any need for this specific risk assessment (Estonian University of Life Sciences, private communication).

In Brazil, the introduction of alien species is forbidden and requires a licence and specific authorisation from the environmental agency (Universidade Federal da Bahia, private communication). Bumblebees, for example, have been shown to escape management, resulting in an invasive species in the wild [35]. Chile participates in the trade of bumblebees for commercial pollination, which is believed to have spread two alien bumblebee species, *Bombus ruderatus* and *Bombus terrestris*, into Argentina, which bans the import of commercial alien bumblebees [2, 35]. Arbetman *et al.* [36] and Maharramov *et al.* [37] found that *Bombus terrestris* introduced the pathogenic protozoan *Apicystis bombi* into the native species *Bombus dahlbomii*. Their conclusions were based on *Apicystis bombi* samples from before and after the introduction of *Bombus terrestris*. They found that the pathogen had co-evolved with *Bombus terrestris* and had been one of the causes for the population collapse of *Bombus dahlbomii* [36]. Another cause for the *Bombus dahlbomii* decline is the overlap in visited plant species compared to the two invasive *Bombus* species [38]. Five years after arrival, *Bombus terrestris* has become the most abundant and widespread *Bombus* pollinator. Furthermore, 20 years after their introduction, *Bombus terrestris* and *Bombus ruderatus* had replaced the only native species, *Bombus dahlbomii*, which was formerly the most abundant pollinator in Patagonia [38].

The outperforming of *Bombus dahlbomii* can have detrimental effects on the local pollination networks as the plant–pollinator interactions have changed. The native bumblebee species *Bombus dahlbomii* is a more efficient pollinator for *Alstroemeria aurea*, for example, in terms of pollen quantity and quality deposited per flower visit. Owing to a shift towards *Bombus terrestris* and *Bombus ruderatus*, *Alstroemeria aurea* could suffer from a reduced fruit set and/or quality [32].

In addition to the abovementioned cases in South America, similar cases have appeared in Asia. In 1992, *Bombus terrestris* was introduced into Japan for the pollination of greenhouse crops and, nearly 20 years later, as a result of an adverse ecological impact, the species was banned. Highlighting one such adverse effect, Inoue and Yokoyama

[33] observed the decrease in terms of population size of the two native bumblebee species, *Bombus hypocrita sapporoensis* and *Bombus diversus tersatus,* simultaneously with the increase of *Bombus terrestris.* The authors provided circumstantial evidence for the competitive exclusion of the two native bumblebees. Tsuchida *et al.* [39] reviewed the hybrid production between the invasive *Bombus terrestris* and the native *Bombus ignitus* and *Bombus hypocrita sapporoensis* in Japan. Owing to interspecific mating, reproduction of the native bumblebee species *Bombus ignitus* and *Bombus hypocrita sapporoensis* is reduced through the production of non-viable eggs [39]. As a result, the replacement of the native bumblebees by *Bombus terrestris* is accelerated.

In 1880, *Bombus terrestris* was introduced into New Zealand for the pollination of red clover, and, merely five years later, populations were found in the wild. Subsequently, the pollinator expanded its territory at a rate of approximately 90 km/year, becoming the main pollinating bumblebee [32]. *Bombus ruderatus*, which was introduced around the same period, became ubiquitous as well, although in some areas the *Bombus ruderatus* population has declined, especially after the *Bombus terrestris* introduction [32]. International companies such as Biobest and BVT should therefore pay attention to the local situation. In fact, BVT claims implementation to be fairly easy as European bees are relatively common around the world or, if not, there is typically some form of commercial pollinator available (BVT, personal communication). The company only works with approved or available pollinators in every country. However, further work must be done as international companies would need local partners in every country. In some countries (e.g. Estonia) no local bumblebee producers are present and rearing the local breeds for every country could be impractical (Estonian University of Life Sciences, personal communication). Furthermore, the International Air Transport Association's Live Animal Regulations include guidance on packaging honeybees and bumblebees but no other pollinators. This could be a barrier to a developing market for some solitary or stingless bees, for example, which are not specifically mentioned in the guidance.

Legislation, registration and the potential risk of biocontrol agents

Several microorganisms including viruses, bacteria and fungi have been suggested as possible microbial BCAs, and at least a dozen have already been tested. Registration requirements for microbial BCAs differ among countries. In the United States of America (USA), the Environmental Protection Agency (EPA) categorises microbial BCAs as biopesticides that must meet the standards of the Federal Insecticide, Fungicide, and Rodenticide Act, which prevents them from 'causing unreasonable adverse effects on human and environmental health' [40]. This is also the case for the BCAs used in entomovectoring. For example, BVT only works with BCAs that are authorised by the respective country (BVT, private communication). In its legislation, the EPA encourages the development of biopesticides via shorter registration procedures and lower registration fees [40]. In the European Union, the approval of microbial BCAs is done at strain level by the European Commission. However, compared to chemical substances, guidance on risk assessment of microbial active substances is lacking. Also in the USA, guidelines on chemical substances are more extensive than their microbial alternatives. Despite the lack of general guidance on the evaluation of microbial BCAs in the European Union, the Organisation for Economic Co-operation and Development's guidelines on the environmental safety evaluation of microbial BCAs are generally followed [40]. Generally, the use of active microorganisms carries risk because of their potential to infect and multiply in host organisms and to produce toxic secondary metabolites upon contact with non-target organisms. These risks should be assessed unless non-target organism exposure can be disproved [40].

In terms of the entomovectoring system in particular, harm to the vector is another focus of attention [7]. For example, *Bacillus subtilis* QST713, the active component of Serenade, has a lethal concentration of 50% (LC_{50}) of 5,663 parts per million (ppm) for honeybees. However, honeybees directly sprayed would be exposed to approximately 8,000 ppm, clearly above the LC_{50}. Nevertheless, the EPA [41] found no adverse effects on pollinators, pollination or yields during field trials where Serenade was transported by honeybees. Mommaerts *et al.* [42] investigated the effect of AQ10 (*Ampelomyces quisqualis*), Binab-T-vector (*Hypocrea parapilulifera* + *Trichoderma atroviride*, 1/1), Prestop-Mix (*Gliocladium catenulatum* J1446), Serenade (*Bacillus subtilis* QST713), Trianum-P (*Trichoderma harzianum* T22), Botanigard (*Beauveria bassiana* GHA) and Granupom (*Cydia pomonella* granulovirus), comprising five biofungicides and two bioinsecticides, on *Bombus terrestris*. Apart from Botanigard and Serenade, the BCAs had no adverse effects on the workers of *Bombus terrestris* [42]. Topical contact with Botanigard at its maximum field recommended concentration (MFRC) caused 92% mortality after 11 weeks, while 1/10 MFRC killed 46% of exposed workers. Topical contact and oral delivery of Serenade resulted in 88% and 100% worker mortality, respectively. Furthermore, upon treatment with their respective MFRC, nests were evaluated for sublethal effects over 11 weeks. Botanigard and Serenade gave rise to a significant decrease in drone production. Finally, Botanigard negatively affected the insect's foraging behaviour when orally delivered at its MFRC.

Additional topical tests are essential to determine a system that does not harm the vector, specifically when the vector comes in direct contact with the BCA. Furthermore, because of the increased globalisation of these emerging options,

international guidelines should augment the national ones facilitating commercialisation.

Conclusions

Entomovectoring combines the value of pollination with useful antipathogenic properties. In doing so, it further increases the international market for the vector species used. Although, in contrast to regular pollinator transport, no extra management is needed during vector transport, misperceptions about the purpose, content and risks may occur as BCAs and vectors are packed separately. It is important to provide a detailed description of the purpose for both the vector and the BCAs. Honeybees and bumblebees are the main pollinators for the entomovectoring system, and both are already treated as special cases in international transport. However, the abovementioned historic cases presenting pollinators that are also frequently used in the entomovectoring system confirm the value of developing future systems using local species, where feasible.

Despite the practical challenges in countries lacking local rearing facilities or relevant research, use of local species/populations for future entomovectoring eliminates the transport issues described in this thematic issue. Furthermore, these investments may help to protect pollinator biodiversity locally and also lead to higher efficiencies in the system. In the meantime, entomovectoring needs international guidelines for the registration of BCAs in order to deliver benefits sooner, including testing for (sub)lethal effects upon topical exposure.

Utilisation d'abeilles diverses en tant que vecteurs pour l'entomovection, à des fins d'amélioration de la pollinisation et de protection des cultures : pratiques actuelles, exemples d'utilisation et regard critique sur le transport

J. Temmermans & G. Smagghe

Résumé

Depuis plusieurs années, les abeilles mellifères, les bourdons et d'autres insectes sont commercialisés à des fins de pollinisation, parallèlement à l'utilisation désormais largement répandue de micro-organismes pour la lutte biologique contre les nuisibles. Les pollinisateurs et diverses formulations d'agents microbiens destinés à la lutte contre les nuisibles font régulièrement l'objet de transports internationaux à grande échelle. Une approche innovante mise au point récemment consiste à utiliser les abeilles en tant que vecteurs d'agents microbiens, en inoculant ces derniers sur la surface du pollinisateur par le biais de diffuseurs disposés dans les ruches modifiées à cette fin. Cette innovation amplifie le marché de ces produits et se traduit par un meilleur rendement. La réussite d'un système d'entomovection passe par la sélection judicieuse du vecteur pollinisateur en fonction des cultures à protéger et du site, sur la base de plusieurs critères, et par son utilisation en association avec un agent antimicrobien autorisé. À l'heure actuelle, les pollinisateurs et les agents microbiens sont conditionnés séparément et leur assemblage est effectué sur le site même d'utilisation. L'approvisionnement local en pollinisateurs au sein du système réduit la nécessité de transporter ces insectes vivants sur de longues distances et pourrait améliorer l'efficacité du système au moyen d'adaptations locales ; néanmoins, cela retardera la mise en œuvre du système et l'obtention de résultats bénéfiques, le temps que la recherche conduite dans les sites ou les pays utilisateurs atteigne le niveau des avancées déjà enregistrées dans un petit nombre d'autres pays. En attendant, des directives claires en faveur des systèmes innovants basés sur l'utilisation d'insectes vivants pourraient œuvrer à l'appui d'une croissance prometteuse de la production alimentaire.

Mots-clés

Abeilles – Contrôle des agents microbiens – Entomovection – Pollinisation – Protection des cultures – Systèmes biologiques innovants.

Sistemas entomovectoriales con empleo de distintos ápidos como vector para mejorar la polinización y el control para la protección de cultivos: métodos actuales, ejemplos de uso y visión crítica del transporte

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Resumen

El uso comercial de abejas melíferas, abejorros y otros insectos con fines de polinización tiene ya muchos años. También está muy extendido el empleo de agentes microbianos de control biológico como método de lucha contra las plagas. El transporte internacional y a gran escala de polinizadores y de ciertas formulaciones de plaguicidas microbianos es algo muy habitual. Ahora se ha concebido un novedoso planteamiento en el que se utilizan ápidos como vectores de agentes microbianos. Para ello, se impregna de estos agentes la superficie del insecto polinizador empleando dispensadores en colmenas modificadas. Tal innovación amplía el mercado de estos productos y de-para mejores cosechas. Para que un sistema entomovectorial sea eficaz es preciso seleccionar el vector polinizador que mejor encaje con el cultivo y las condiciones locales atendiendo a diversos criterios y combinarlo con un agente microbiano registrado. Actualmente, los polinizadores y los agentes microbianos se embalan por separado y se combinan solo en el punto de aplicación. El aprovisionamiento del polinizador del sistema a partir de fuentes locales hace menos necesarios los envíos a larga distancia de estos insectos vivos y puede ofrecer más eficacia gracias al mayor grado de adaptación del insecto a las condiciones locales. Sin embargo, ello retrasará el uso del sistema y la obtención de los consiguientes beneficios, pues habrá que esperar a que en cada lugar o país se haya llevado el procedimiento tan lejos como se ha hecho hasta ahora en unos pocos países. Mientras tanto, la existencia de claras pautas sobre estos innovadores sistemas en que se emplean insectos vivos podría traducirse en un prometedor aumento de la producción alimentaria.

Palabras clave

Agente microbiano de control – Ápidos – Polinización – Protección de cultivos – Sistema entomovectorial – Sistemas biológicos innovadores.

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